CHAPTER

25

Esthetics and Smile Designing

The clinical practice of dentistry involves morphological, functional and esthetical components. Esthetics is an intellectual phenomenon, which deals with scientific knowledge, principles and perception of essential and natural beauty in nature and art. Esthetic dentistry is the "*art and science of dentistry applied to create or enhance the beauty of an individual within functional and physiological limits*". It involves the skills and techniques used to improve the symmetry of the teeth and face to enhance the appearance as well as functions of teeth, oral cavity and the face.

The history of esthetic concepts dates back to more than 4000 years. Over the years, newer materials and techniques have revolutionized the dental restorations, creating esthetically pleasing facial expressions.

PRINCIPLES OF ESTHETICS

The basic principles of esthetics are:

I. Composition
II. Proportion
III. Illusion
IV. Color
V. Translucency

I. Composition

Composition, the act of combining elements, can be defined as "*the relation that exists between the teeth (when illuminated), due to attributes of contrasts in color, lines and texture*". Any composition can be esthetic or non-esthetic depending on how these elements are combined.

Various attributes of elements of composition that impart the esthetic value are:

a. ***Unity or oneness:*** A dentition appears pleasing when teeth are arranged in repetitions of their shape (outline form), line (axial inclination and position) and color. This repetitive fashion in right and left sides in the oral cavity gives unity to the composition.
b. ***Subtle variation:*** The unity of the composition is not static but dynamic; nature never provides absolute bilateral symmetry. There exist subtle variations in right and left halves of the dentitions, such that dynamic unity can be perceived.
c. ***Dominance:*** Dominance implies when a strong structure in the center is surrounded by well-demarcated structures. The centrally placed structure dominates the visual perception.
   For example, when numerous anterior natural dentitions are viewed, then one feature that will be dominant in all is maxillary central incisor (*Color, shape and size are the factors that control the dominance*).
d. ***Contrast:*** Contrast visualizes various elements of a composition. The eye can differentiate the parts of an object due to contrast of colors, lines, patterns, etc.
e. ***Symmetry:*** It is the regularity of arrangement of objects either in horizontal/vertical direction, or from a central point to either side, like a mirror image (radiating symmetry). In a dentofacial composition, radiating symmetry of the teeth is esthetically more appealing.

II. Proportion

The term 'proportion' is used to ascertain mathematical representation to the relationship of the various units in a composition.

In natural dentition, though each individual tooth has a different shape and size, they are related to one another and the entire dental composition relates to face in a certain proportion, that must be pleasing and harmonious. The proportion in esthetics is the key tool in shaping the dental and facial components.

The routinely used ratios applied in dentistry are:

a. Golden proportion
b. Golden percentage or golden mean
c. Red proportion
d. Esthetic gauges

a. Golden Proportion

The golden proportion, also named as 'divine proportion', represents the most agreeable proportion

of two measurements. This ratio is approximately 1.61803:1; that is, the smaller section is about 62% the size of the larger. The uniqueness of this ratio is that the ratio of the smaller part to the larger part is the same as the ratio of the larger part to the whole. It is rounded to a mathematical figure—1.618 (labeled as phi, using Greek symbol).

- Natural law describes how our eyes evaluate a pleasing proportion.
- Applied in arts, architecture and anatomy.
- Display maximum beauty and ultimate efficiency in function.

This mathematical phenomenon is also called the 'magic numbers'. To create the series, begin with the simple addition equation of 0 + 1 = 1. The second number to be added in that equation, 1, and the sum of that equation, also 1, are added together, in that order, to create the next line (1 + 1 = 2). The sums of each line—1, 2, 3, 5, 8, 13, etc., are the Fibonacci series.

0 + 1 = 1
1 + 1 = 2
1 + 2 = 3
2 + 3 = 5
3 + 5 = 8
5 + 8 = 13, etc.

This pattern is not a simple arithmetic progression; it follows an exponential law that can be applied to esthetic dentistry. This progression is most pleasing to the senses and provides a unique relationship. The Fibonacci numbers, after the thirteenth in the series, increase in an unchanging proportion of 1.0 to 1.61803. Therefore, in that part of the series, the Fibonacci numbers are golden to their neighbors.

Divide:

0 / 1 = 0	34 / 21 = 1.61904
1 / 1 = 1	55 / 34 = 1.6176
2 / 1 = 2	89 / 55 = 1.61818
3 / 2 = 1.5	144 / 89 = 1.61797
5 / 3 = 1.666	233 / 144 = 1.61805
8 / 5 = 1.6	377 / 233 = 1.61802
13 / 8 = 1.625	610 / 377 = 1.61803
21 / 13 = 1.615	987 / 610 = 1.61803

Clinical implications

- The upper central incisor has a golden ratio to the lower central incisor, if the lower central incisor is taken as a starting reference.
- The total width of both lower centrals is golden to that of the upper incisors.
- Similar series of proportions lies with the widths of all four upper incisors to lower four incisors.
- Width of lower incisor segment to the upper canine width appears in golden ratio.
- Similar relationship is present from the distal of the lower canines to the buccal groove on the upper molar.
- Width of the maxillary lateral incisor, as viewed from the front, should be in golden proportion to the width of maxillary central incisor. The maxillary lateral incisor should be 62% of the width of the maxillary central incisor. Similarly, the width of maxillary canine, as viewed from front should be 62% of the width of lateral incisor (Fig. 25.1).

b. *Golden Percentage or Golden Mean*

The Golden percentage/mean implies that the width of the maxillary central incisor should be 25% of the total width from distal of maxillary canine on one side to the distal of canine on the contralateral side. Each maxillary lateral incisor should be 15%; and each maxillary canine 10% of the intercanine distance, as viewed from the front (Fig. 25.2).

c. *Red Proportion*

The red proportion implies the proportion of successive widths of the maxillary teeth should remain constant, progressing distally. The width of the lateral incisor is reduced by a selected percentage from the width of the central incisor, and the width of each tooth distally is reduced by the same percentage from its mesial

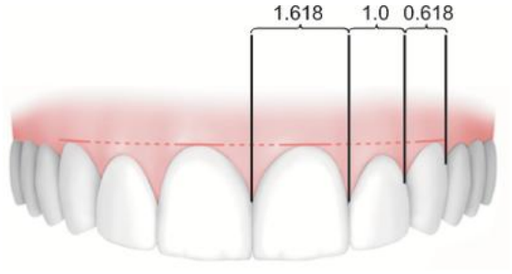


Fig. 25.1: Golden proportion

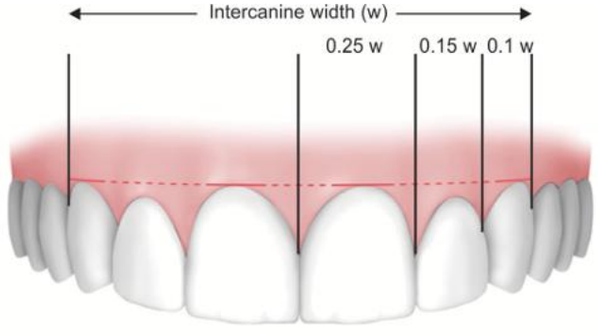


Fig. 25.2: Golden mean

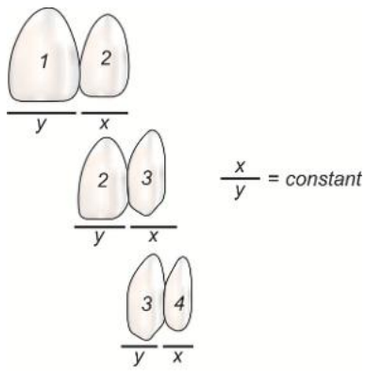


Fig. 25.3: Red proportion

tooth. The 70% red proportion has been recommended for normal length; when using 70% red proportion, the width of the maxillary lateral incisor is 70% of the maxillary central incisor, and the maxillary canine is 70% of the width of the resulting lateral incisor (Fig. 25.3)

- 70% red proportion is recommended for normal length.
- 62% red proportion is recommended for very small teeth.
- 80% red proportion is recommended for small teeth.

d. *Esthetic Gauges*

A series of gauges are available for intraoral analysis. These gauges analyse tooth width, length and gingival discrepancies. The gauges are used as a guide between clinician and technician.

III. Illusion

The art of creating illusion consists of changing perception, to cause an object to appear different from what it actually is. Teeth can be made to appear smaller, larger, wider, narrower, shorter, longer, younger, older, masculine or feminine. *For example:*

a. *Illusion achieved from the principle of gradation of size:* The central incisor closest to the viewer appears the largest; whereas, the premolar appears smaller because they are farther away from the viewer, even both teeth may have same occlusogingival length.

b. *Illusion achieved from light/illusion of transillumination:* An object exhibits two dimensions, that is, length and width when subjected to light. True natural light is multi-directional and on striking the surface of the object, also reveals texture and shadows (This is third dimension of depth).
   - The horizontal lines illuminated under light make the object wider and vertical lines make the object appear longer (principle of line).
   - Shadows created as light strikes the facial surface of the tooth begin at the transitional line angles; these shadows delineate the boundaries of the facial form. Thus the apparent face is that portion of the face that is visible to the observer from any single view. When maxillary anterior teeth are viewed from the front, only the mesial half of canine is visible; however, the entire incisor is visible.

IV. Color

Teeth are composed of multitude of colors. A gradation of color can be perceived from gingival to incisal, with the gingival area being darker because of thinner enamel. The use of different shades of restorative material may be required to esthetically restore a tooth.

V. Translucency

Translucency implies as to how deeply light penetrates into the tooth before it gets reflected outward. Normally light penetrates through the enamel into dentin before it gets reflected outward. This provides vital characteristic of normal, unrestored tooth. Shallow penetration often results in loss of esthetics. *Illusion of translucency* can also be created to enhance the realism of restoration (addition of tints or opacifiers).

ESTHETIC SMILE DESIGNING

The following features are thoroughly analyzed before sequential planning of smile designing:

A. Facial Analysis

a. *Facial height:* The face height is divided into three equal thirds (Fig. 25.4):
   - From forehead to brow line

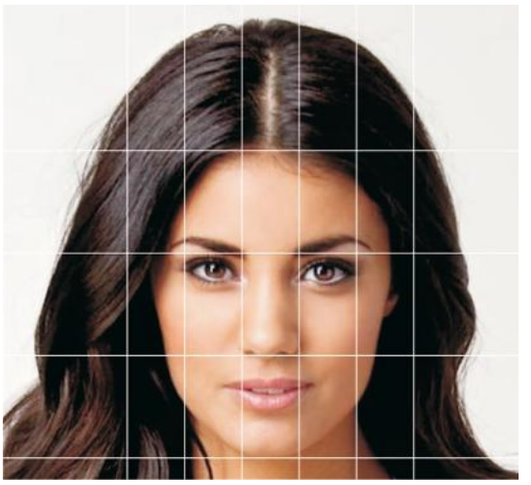


Fig. 25.4: Facial analysis



Fig. 25.5: Lip analysis: **(A)** Gummy smile; **(B)** Adequate exposure of incisors; **(C)** improved smile

- From brow line to the base of the nose
- From base of the nose to base of the chin.
- Full face is divided into two parts; eyes being in midline.
- Face from base of nose to the chin is divided into two parts; upper lip is one-third and the lower-lip and chin two-thirds of it.

b. *Facial width:* The width of face is the width of five 'eyes' (Fig. 25.4). The distance between eyebrows and the chin should be equal to the width of face.

c. *Facial profiles:* Lateral profile can be straight, convex, or concave (A cephalometric analysis of the head in frontal and lateral views is useful in determining bony relationships of the face and the mandible, and their relationship to the teeth in the alveolar bone). The shape of face from frontal aspect can be square, tapering, ovoid and square-tapering.

d. The facial features related to gender and age involve the soft tissues and include the texture, complexion, and tissue integrity of the epithelial tissues.

e. Facial features having impact on the dental-facial composition are those that relate the interpupillary plane with the commissure line and the occlusal plane.

B. Lip Analysis

a. *Lip morphology:* Width, fullness and symmetry.

b. *Lip line:* The inferior border of the lip as it relates to the teeth and gingival tissues is called the *lip line*

- Average lip line exposes the maxillary teeth and only the interdental papillae.
- A high lip line exposes the teeth in full display as well as gingival tissues above the gingival margins (gummy smile).
- A low lip line displays no gingival tissues when smiling.

Mostly, the lip line is acceptable, if it is within a range of 2.0 mm apical to the height of the gingiva on the maxillary central incisors (Fig. 25.5A to C).

C. Incisal Display (Incisal Length and Edge Position)

The incisal display refers to the amount of visible tooth displayed when the lips and lower jaw are in the rest position.

The important parameters for establishing incisal display are:

i. *Degree of tooth display:* It is hypothesized that 3.5 mm of incisal-third of maxillary central incisors should be visible.

ii. *Incisal embrasures:* Incisal embrasures from central-central to central-lateral and lateral-canine is specified for males and females (Fig. 25.6A, B). Open incisal embrasures characterize feminine smile; whereas, less open incisal embrasures characterize masculine smile.

iii. *Phonetics and sound:* Phonetics and sounds affect the placement of incisal teeth vis-à-vis tongue and lips. Phonetics, such as 'M' (lips return to their normal rest position); 'E' (incisal edge positioned half way between upper and lower lip); 'S' (mandibular incisors are positioned 1.0 mm behind and 1.0 mm below the maxillary incisal edge).

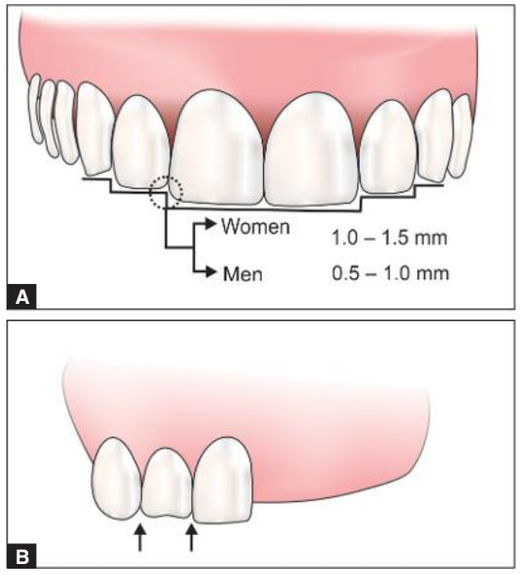


Fig. 25.6: Incisal edge position: **(A)** Open incisal embrasure (female); **(B)** Less open incisal embrasure (male)

With age, the amount of incisal display of the maxillary central incisors diminishes and the amount of incisal display of the mandibular central incisors increases (Amount of incisal display is important in a young smile).

D. Gingival Tissue Analysis

a. ***Gingival contour:*** The gingival contours should be symmetric and the marginal gingival tissues of the maxillary anterior teeth should be located along a horizontal line extending from canine-to-canine (Laterals may be slightly short of that line). It is also acceptable to have the gingival height of all six anteriors equal in gingival height on the same plane (Fig. 25.7).
Ideally the papillary contour should be pointed and should fill the interdental spaces to the contact point. An unfilled interdental space creates an unwanted black interdental triangle in the gingival embrasure and makes a smile less attractive.

b. ***Zenith point:*** The gingival zenith point is the most apical point of the gingival tissues along the long axis of the tooth (Fig. 25.8). Zenith point can enhance:
- Preparation of tooth axis
- Length of the tooth
- Gingival shapes

Importance of zenith point
- When closing diastemas or changing the mesial or distal tilt position of the tooth (These are examples of moving zenith points horizontally.)
- In case where tooth needs to be shown longer or more taper at the gingival 1/3rd, zenith point can move apically.

c. ***Biological width:*** Biological width is defined as *"the combination of junctional epithelium and connective tissue elements of dentogingival unit that occupies the space between base of the gingival crevice and the alveolar crest; the total dimension would be 2.04 mm"*. It was termed as 'biologic zone' by Kois (3.0 mm in healthy normal gingiva).

i. Epithelial compartment
- Gingival epithelium
- Sulcular epithelium
- Sulcus depth—0.69 mm
- Functional epithelium—0.97 mm

ii. Connective tissue compartment
- Represents fibrous attachment of the gingiva to the hard tissue wall and support to the epithelium of the dentogingival junction (Fig. 25.9).

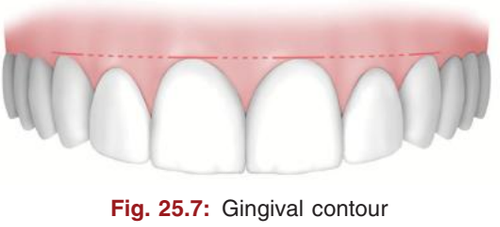


Fig. 25.7: Gingival contour

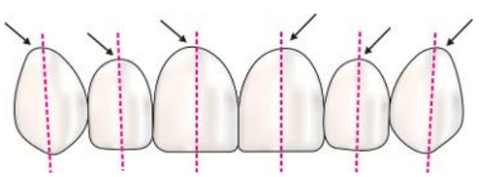
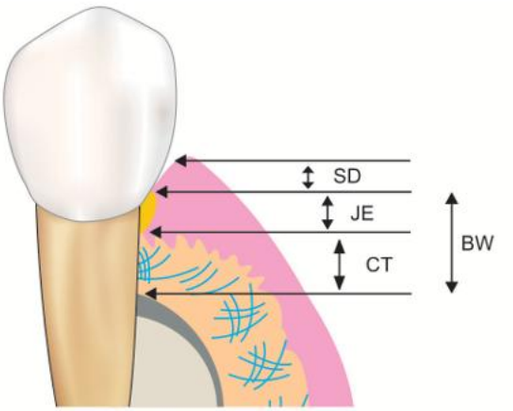


Fig. 25.8: Zenith point



SD : Sulcular depth
JE : Junctional epithelium
CT : Connective tissue
BW : Biological width

Fig. 25.9: Biological width (biologic zone)

E. Smile Line

The smile line can be defined as *'an imaginary line drawn along the incisal edges of the maxillary anterior teeth'* (Fig. 25.10).
- In an ideal tooth arrangement, smile line should coincide or follow the curvature of the lower lip while smiling.
- The central incisors are slightly longer than the cuspids (another reference).



Fig. 25.10: Smile line

- In case the central incisors appear shorter than the cuspids along the incisal plane, it creates an aged or worn appearance (reverse smile line).

F. Midline

The midline refers to a vertical line formed by the contact of the maxillary central incisors. The midline should be perpendicular to the incisal plane and parallel or coincident to the midline of the face (Fig. 25.11).

Several anatomical landmarks can be useful guides to assess the midline of the face as it relates to dental midline. These include:

- Midline of the nose
- Interpupillary plane
- Philtrum
- Midline of chin

(The philtrum of the lip is considered to be one of the most accurate anatomical guides as it is always in the center of the face.)

Minor discrepancies between facial and dental midline are acceptable (maximum discrepancy allowed is 2.0 mm). Location and alignment are the two important features of midline.

- Midline should be parallel to the long axis of the face.
- Midline should be perpendicular to the incisal plane.
- Midline should be just straight down the papilla.

Maxillary and mandibular midlines may not coincide (*Mandibular midline is not used as reference point for establishing maxillary midline*). Maxillary midline is more effective in smile as compared to mandibular midline (not visible during smile).

G. Axial Inclination

Axial inclination refers to the degree of tipping any plane of reference (Maxillary central incisors positioned vertically/slight labially; lateral incisors' incisal edge inclined, labially and cervical tucked in; canines' tips are angled lingually and cervical labially).

The interproximal area follows 50:40:30 rule in reference to maxillary central incisor (Increasing the contact area helps to create the illusion of longer teeth and minimize the black triangles) (Fig. 25.12A, B).

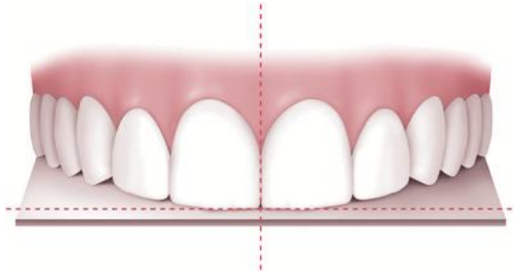


Fig. 25.11: Midline

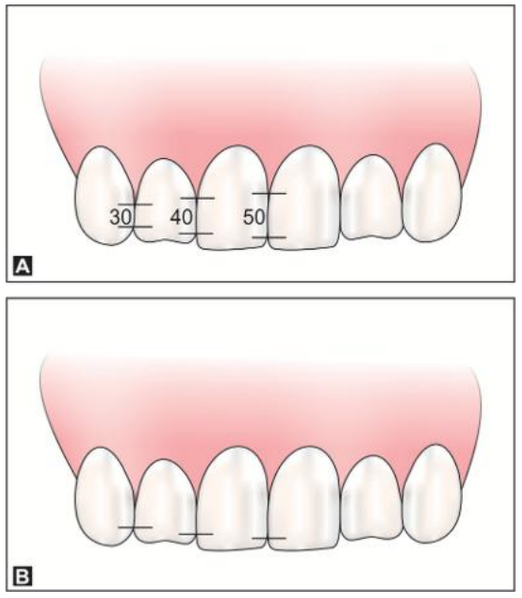


Fig. 25.12: Axial inclination: **(A)** Axial inclination (50:40:30 rule); **(B)** From central to canine, contact moves apically

The axial inclination of the anterior teeth tends to incline mesially toward the midline and becomes more pronounced from the central incisors to the canines:

i. The axial inclination of the posterior teeth from the frontal view exhibits the same mesial inclination toward the midline as the cuspid.
ii. This creates a natural visual gradation, making the teeth appear to diminish in size as they progress posteriorly.

H. Buccal Corridor

While smiling, the areas between the corners of the mouth and the buccal surfaces of the posterior maxillary teeth form a space known as the 'buccal corridor'. The more pronounced this space becomes, the more these posterior teeth are concealed, restricting the full breadth of the smile. A symmetric buccal corridor is an important element of an esthetic smile. In these posterior segments, the artistic perception of esthetics can be used to alter the inclinations to produce an enhanced esthetic effect (Fig. 25.13).

I. Anatomy of Anterior Teeth

The factors guiding individual tooth dimensions are:

- Maxillary central incisor, create central dominance, 10–11 mm long and 80% wide of their length.
- Maxillary lateral incisors, usually asymmetrical and influence gender characterization.



Fig. 25.13: Buccal corridor

- Maxillary canine, depicts the personality characterization; aggressive in males and delicate in females.
- Central incisors are wider than laterals by 2.0–3.0 mm; canine is wider than lateral by 1.0–1.5 mm; canine/central are longer than lateral by 1.0–1.5 mm.

The important features of anatomy of anterior teeth are as follows:

a. *Labial contour:* The labial contour of anterior teeth should exhibit three planes when viewed from a lateral profile (three reflections of light) (Fig. 25.14). The shape could be convex, square or fan-shaped, when viewed from front.
b. *Incisal embrasure:* An open space is present between the proximal surfaces of anterior teeth from the contact points to incisal surface. This area is called an 'incisal embrasure'. The incisal embrasure spaces display a natural and progressive increase in depth from the central incisor to the canine (Fig. 25.6A).
c. *Texture:* The characterization of the tooth surface is a function of anatomical grooves, facets, prominences along with Perikymata, stippling and rippling that may affect the enamel surfaces.

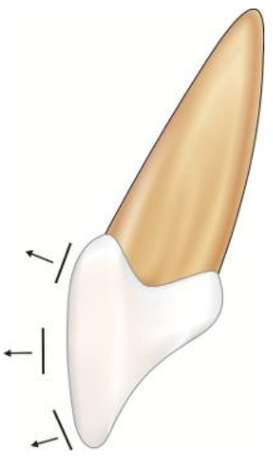


Fig. 25.14: Labial contour reflects light in three planes

d. *Mesiodistal width:* This dimension is much more critical dimension than incisogingival length for anterior tooth placement.
e. *Incisogingival height:* The incisal edge of the maxillary central incisor is an important determinant in creation of a smile. It serves to determine the proper tooth proportion and gingival level. Position of the incisal edge acts as the parameter upon which rest of the treatment is built. Average anatomic crown length for maxillary central incisors is 10.4–11.2 mm. The ideal ratio of incisogingival height and mesiodistal width of central incisors is shown in Fig. 25.15. Ideally the mesiodistal width of the tooth must be 80% of the incisogingival height of the tooth.

The modification (elongation/shortening) of this ratio is indicated in certain cases to achieve the requisite smile.

The indications are:

i. Elongation of the incisal edge:
   - Incisal wear
   - Inadequate tooth display
   - Displeasing tooth or crown proportion

ii. Shortening of the incisal edge:
   - To correct excessive tooth display
   - Displeasing tooth/crown proportion
   - To compensate for unesthetic elongation created by periodontal recession.

Table 25.1 summarizes features of smile designing.

TREATMENT PLANNING

The treatment planning for the esthetic restoration modalities involves physical analysis of the stomatognathic system and the psychology of the patient as well. The following features should be analyzed initially:

- Thorough knowledge of esthetic principles
- Patients' needs and aspirations
- Esthetic perception of the patient
- Psychology of the patient
- Components of dentofacial complex
- Physiology of occlusion and mastication
- Affordability



Fig. 25.15: Ratio of mesiodistal width to incisogingival height of maxillary central incisor

Treatment planning starts with evaluation of entire stomatognathic system; however, following three components are mandatory for planning esthetic restorations.

- Evaluation of the teeth and arch
- Evaluation of the periodontal status
- Analysis of facial structures

Various esthetic issues and the preferred treatment modalities are summarized in Table 25.2.

The commonly used treatment modalities are:

1. Cosmetic Contouring

Cosmetic contouring is the reshaping of the natural teeth for esthetic purposes. Such reshaping does not merely consist of filing and leveling the incisal edges, but it involves shaping the mesial, distal, labial and lingual surfaces as well. Cosmetic reshaping provides excellent results in many situations where other

Table 25.1: Smile designing

Dominance of central incisors	i. Smile arc (maxillary incisal edges contouring the lower lip) (convex or curved)	• Incisal edge of maxillary central incisor should be below the tip of canine • Step between central and lateral incisor be in the range of 1.0–1.5 mm for females and 0.5–1.0 mm for males • Arch looks provide younger smile looks; whereas, plane/inverted provide older smile looks.
	ii. Symmetry/ratio (maxillary anteriors)	• Width to height ratio be 75–80% [less than 75% (long pattern) may suit females and more than 80% (wider pattern) suit males] • Incisal edge symmetry of maxillary anterior teeth
	iii. Proportions (central to canines)	• Wider incisors may look more esthetics up to range of 70% golden proportions • Narrow lateral incisors are unesthetic • Diastema of any size is unesthetic
	iv. Midline	• Midline deviation more than 2.0 mm is critical; need correction
	v. Angulation of teeth (parallelism in central incisors and interpapillary line)	• Incisal angulation from frontal and lateral profiles affects esthetics (tooth angulation is more relevant than midline)
	vi. Color	• Bleaching is recommended to improve color • Contacts should be adequate to correct potential black spaces
Dominance of gingival architect	i. Gingival contour (gingival apex is located in the center of crown or slight distally)	• Gingival margins of central incisors be at level or slight below the canines • Gingival margins of lateral incisors be slightly below the central incisors
	ii. Gingival exposure (within 3.0 mm is acceptable; more than 3.0 mm may be unesthetic) smile line can be high, low or medium	• Intrusion of incisors is not preferred • Gingival exposure more than 3.0 mm can be treated accordingly
	iii. Lip volume (voluminous lips considered standard of beauty)	• Artificial lip filling can be performed in case of thin lips
	iv. Buccal corridor (bilateral space between vestibular surface of maxillary posterior teeth and lip commissure while smiling	• Inward inclination of lips are avoided (retraction of maxillary anteriors may lead to inward incline lips) • Avoid wider buccal corridor (intermediate buccal corridor is more pleasing)

Table 25.2: Treatment options for various esthetic issues

Esthetic issues	*Treatment modalities*
Tooth morphology and alignment	• Bleaching/microabrasion • Recontouring • Composite resin restorations • Composite/ceramic veneers • Esthetic inlays, onlays and crowns
Stomatognathic system	• Orthodontic treatment • Orthognathic surgery • Post-core restorations
Color issues	• Bleaching • Veneers • Crowns
Traumatic injuries	• Splinting • Surgical corrections • Periodontal therapy • Veneers/crowns
Gingival/ periodontal issues	• Gingival recontouring • Electrosurgery • Gingival grafting • Periodontal therapy • Frenectomy
Missing teeth	• Dentures/bridges • Implants

esthetic procedures are practically difficult. It is a rapid procedure that provides long-lasting results and is nearly painless.

Photographs, study models, computer-aided esthetic designing, mock-up casts and line drawings are helpful in treatment planning. Feminine characteristics of esthetically pleasing anterior teeth include narrow teeth, rounded incisal angles, prominent or open incisal embrasure. Masculine characteristics include, wider incisors, sharp incisal angles or minimally open incisal embrasures.

Indications

- Alteration of tooth structure
- Correction of developmental anomalies
- Minor orthodontic problems
- Removal of stains and other discolorations
- Minor periodontal problems
- Bruxism
- Minor worn off incisal edges
- Minor facture of tooth enamel

Contraindications

- Large pulp chamber
- Thin enamel
- Deep pigmented stains
- Large anterior restorations
- Susceptibility to caries
- Excessive anterior crowding/occlusal disharmony

Procedure

- Protrusive occlusion should be evaluated prior to initiation of treatment.
- Reshaping is restricted to enamel only.
- Diamond burs and abrasive discs are used to carry out the necessary changes.
- Marking outline of the change enables patient and clinician to envision the potential improvement.
- The reshaped surfaces should be finished and polished to ensure long-term results.

2. Diastema Closure

The presence of a space, or diastema, between anterior teeth is a common feature of adult dentitions. The space usually distorts a pleasing smile; the observer concentrate on the diastema, rather than overall dental composition.

The etiological factors leading to diastema are:

i. *Developmental:* Microdontia, missing laterals, mesiodens, macroglossia, fibrous frenum
ii. *Pathological:* Midline cysts, tumors and local periodontal problem
iii. *Neuromuscular:* Oral habits, such as tongue thrusting, swallowing or abnormal pressure during rest.

Many forms of management can be used for diastema closure, such as porcelain laminate/veneers, composite veneers, and full veneer crowns (Fig. 25.16A, B). A comparison between direct composite veneer and porcelain veneer is enlisted in Table 25.3.

Procedure (Direct Composite)

1. Clean teeth to be treated and the adjacent teeth. Select shade under appropriate light conditions.
2. The mesio-distal width of teeth before restoration is measured with a vernier caliper and recorded. The probable width after restoration should also be noted.
3. Apply rubber dam.
4. Etch one tooth to be treated while protecting the adjacent tooth with a mylar strip. Apply bonding material, air dry and cure.
5. Build up contour of first tooth using composite of previous selected shade.
6. Sof-Lex disc is used to remove facial , incisal and interproximal excesses. To remove lingual excess, a football-shaped bur can be used.
7. Polish interproximal area with fine and extra-fine Sof-Lex discs.
8. Repeat the similar steps for the adjacent tooth.



Fig. 25.16: Diastema closure: **(A)** Preoperative; **(B)** Post-operative

Table 25.3: Comparison between direct composite veneers and ceramic veneers for diastema closure

Direct composite veneers	*Ceramic veneers*
Advantages	
• Conservative treatment modality • Satisfactory esthetic • Minimal tooth preparation required • Economical • Single visit treatment	• Excellent esthetics • Maximum control over shade, contours and proportions
Disadvantages	
• Needs multiple replacements due to color instability • Effective esthetics only in patients with excellent oral hygiene status	• High cost • Two appointment procedure

9. To maintain the separation between the adjacent teeth, keep the mylar strip in place during placement of composite increments.
10. Contour transitional angles of teeth with disc (evaluate contour from incisal, facial and lingual aspect).
11. Polish the restoration. Verify smoothness of contact area and gingival margin with floss.
12. Remove rubber dam. Check and adjust centric and excursive contacts with articulating paper.
    a. To maintain proper length to width ratios, teeth need to be lengthened in a diastema closure, it is possible to lengthen the anterior teeth either apically, with periodontal procedures, or incisally with a restorative addition.
    b. If the incisal edge position is to be maintained, and there is no periodontal intervention, closing the diastema results in short appearing clinical crowns. Such patients may need aggressive restorative approach. It may be necessary to include 4 or 6 teeth in the restorative plan. The distal surface of the teeth in these cases is reduced with addition to the mesial surface. This approach keeps individual tooth proportion appropriate, while at the same time moves the dental midline to the right position. Gingival zenith point location must be a part of the treatment planning for these cases. The zenith points have to be relocated with either a periodontal procedure or by recontouring the gingival trough in the provisional restorations.

The procedural steps for composite veneer and ceramic veneer techniques are given in Chapters 19 and 21, respectively.

3. Adhesive Restoration

Composite resins are mainly used as adhesive material to modify tooth structure for improving esthetics (filling small pits/defects, restoring incisal edges, adding length to the crown, etc.). The indications, advantages, disadvantages of three major types of composites are summarized in Table 25.4.

The composite containing tints and opaquers are used to improve upon esthetics of restorations. The indications of using tints and opaquers are summarized in Table 25.5.

The necessary techniques involve an understanding of color, adhesive principles, layering to create polychromicity, incisal effects, perfect imperfections and finishing and polishing.

a. Layering Techniques

A thin layer of composite is applied to improve upon the optical, anatomic, and functional characteristics of natural teeth. The operator chooses an enamel replacement material that emulates the missing tooth structure in optical properties and strength (Fig. 25.17A).

Table 25.4: Indications, advantages and disadvantages of the three major categories of composite resins

Classification	*Indications*	*Advantages*	*Disadvantages*
Microhybrid	Restorations requiring deep shade within enamel and dentin	• Provide sufficient strength (less likely to chip off in high-strength area) • Opacity similar to enamel and dentin	• Does not sustain polish for long time
Microfilled	Restorations requiring enamel color and translucency	• Long-term high polish shine • Wear resistance better than microhybrids • Tanslucency similar to enamel	• Lacks strength and can be too translucent
Nanofilled	Anterior and posterior restorations	• Advantages of MicroHybrid and MicroFill	• May hinder reflection of light

Table 25.5: Indications of tints and opaquers

Tints	*Opaquers*
• Match natural tooth structure in polychromicity and maverick colors • Help mask out tooth-restorative interface • Lower the value	• Block dark tooth color in thin layer • Help mask out tooth-restorative interface • Block excessive translucency in Class III and IV restorations • Raise the value

The layering technique can be modified according to esthetic requirements of the patient.

Modification 1: Two shade materials, one for enamel and one for dentin are used with an additional layer of incisal shade material (Fig. 25.17B).

Modification 2: Two shade materials with different chromas are used to replace dentin; one layer for enamel effects along with separate incisal shade material (Fig. 25.17C).

Modification 3: Two shade materials with different chromas are used to replace dentin; two shade materials with different chromas for the enamel and another layer uses incisal shade material.

A characterization layer (tints, etc.) can be placed between the two enamel layers or enamel and dentin layer (Fig. 25.17D).

Advantages

- Easy and efficient to use
- Only one session required
- Better esthetic results
- No laboratory work
- Optimum customization (choice of color, highlighting shape and structure)
- Economical

b. Componeer

Componeers are prepoylmerized, prefabricated nano-hybrid composite enamel shells that combine the advantages of direct composite restoration and the prefabricated veneers (Fig. 25.18A, B). The componeer veneers are available in different sizes and are easy to use for single-tooth reconstruction or complete reconstruction in the anterior region. Componeer can

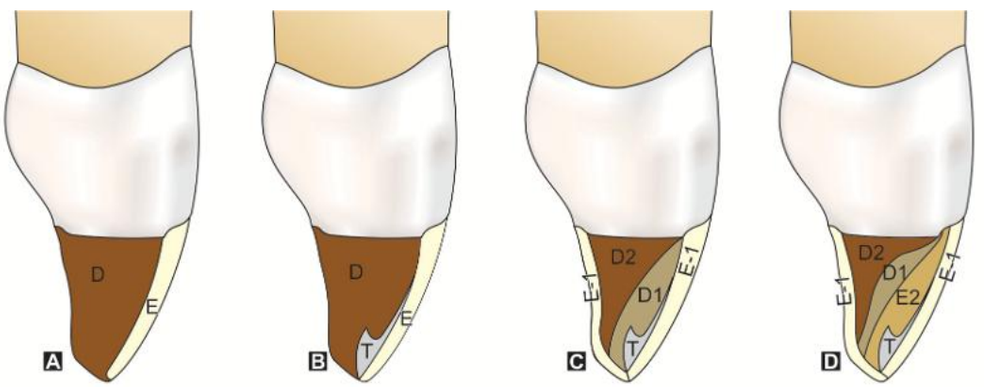


Fig. 25.17: Layering techniques: **(A)** Simple; **(B)** Modification 1; **(C)** Modification 2; **(D)** Modification 3. (E—enamel; D—dentin; T—translucent)



Fig. 25.18: Componeer: **(A)** Preoperative; **(B)** Postoperative

be customized to individual requirements. The extremely thin veneers (as thin as 0.3 mm) allow conservative tooth preparation.

c. *Lumineers*

Lumineers are made from cerinate porcelain, that is sufficiently strong, but much thinner than traditional veneers. Their thickness is comparable to contact lenses; often called contact lenses for teeth. The main advantage of the ultra-thin veneers is the minimal tooth preparation. Lumineers may not require any grinding/cutting of tooth structure for placement. The veneers are adhesively bonded to the facial surface of the teeth, following minimally invasive procedure (Fig. 25.19).

Advantages
• No preparation required; may need minimal adjustment
• Reversible procedure, as no cutting required
• Can be used over fixed crowns
• Excellent esthetics
• Procedure is painless without any postoperative sensitivity

4. Lasers

Minimally invasive laser procedures, with precise restorative planning and technique, can satisfy esthetic and functional parameters. The advantages and disadvantages of lasers are summarized in Table 25.6.



Fig. 25.19: Lumineers placed on anterior teeth

Table 25.6: Advantages and disadvantages of lasers

	Advantages	*Disadvantages*
Soft tissue laser	• No suturing • Little/no bleeding • Painless • Quicker • Atraumatic	• Technique sensitive
Hard tissue laser	• Quicker • More accurate • More comfortable • Better results	• Crack formation • Carbonization of dentin • May damage pulp • Technique sensitive

The types of laser currently used for the esthetic dental practice include argon, carbon dioxide, diode, Er:YAG, and Nd:YAG lasers.

1. ***Argon lasers:*** Used for curing of composite restorations and bonding cements; transillumination of teeth for the purpose of detecting tooth fractures and carious lesions.
2. ***Carbon dioxide lasers:*** Used for cutting and coagulation of soft tissue, which includes gross tissue debulking, frenectomy, gingivoplasty, gingivectomy, etc.
3. ***Diode laser:*** Used for rapid cutting, vaporizing and bacterial reduction adjacent to tooth structure.
4. ***Er:YAG laser:*** Used to cut soft tissue, tooth structure/bone, bleaching of teeth, etching and for treatment of dentin hypersensitivity.
5. ***Nd:YAG laser:*** Used as a hard and soft tissue laser and for selective removal of initial caries with little interaction to the surrounding healthy enamel.

ETHICS AND ESTHETIC DENTISTRY

The term 'cosmetic procedure' refers to elective procedures performed on normal tissues in order to enhance appearance while maintaining functional integrity.

Prior to undertaking any elective, 'cosmetic' dental procedures, it is vital for the operator to discuss the merits and drawbacks of all viable options. It is important that the patient understands what the consequences and limitations of treatment are likely to be, and what the potential failures could entail later in his/her life. Informed consent should be obtained (preferably in writing) and the clinical notes and records should be clearly documented. The operator should be aware of the expectations of patients and be cautious about accepting unrealistic 'cosmetic' expectations. The cosmetic/esthetic dental treatment should be provided, which is minimally destructive and, in the long term, be in the 'best interests' of the patient.

The operator should keep in mind the following ethical issues prior to performing an 'esthetic' treatment.

- The use of false credentials promoted by non-accredited institutions
- Reliance on unproved science to promote treatments
- Exaggeration of clinical skills
- Unnecessary treatment procedures
- Lack of informed consent
- Harmful practices, such as, unnecessary removal of tooth structure and the replacement with heavy prosthesis
- Exposing patients to the unknown risks

All dental professionals own an ethical duty to weigh the benefits and the risks of any procedure, and if the potential harm or risks outweigh the benefits, even patients' requests for treatment should be declined.

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